1. selects and removes the two elements in the list with the smallest frequency
2. then creates a new node by combining them (summing the two frequencies)
3. and finally adds back the new node to the list.

While the tree itself is not a heap, a key step of the algorithm is based on efficiently retrieving the smallest elements in the list, as well as efficiently add new elements to the list. You probably have guessed by now that, once again, that's where heaps come to the rescue.

Let's take a look at the algorithm itself, in listing 2.14.

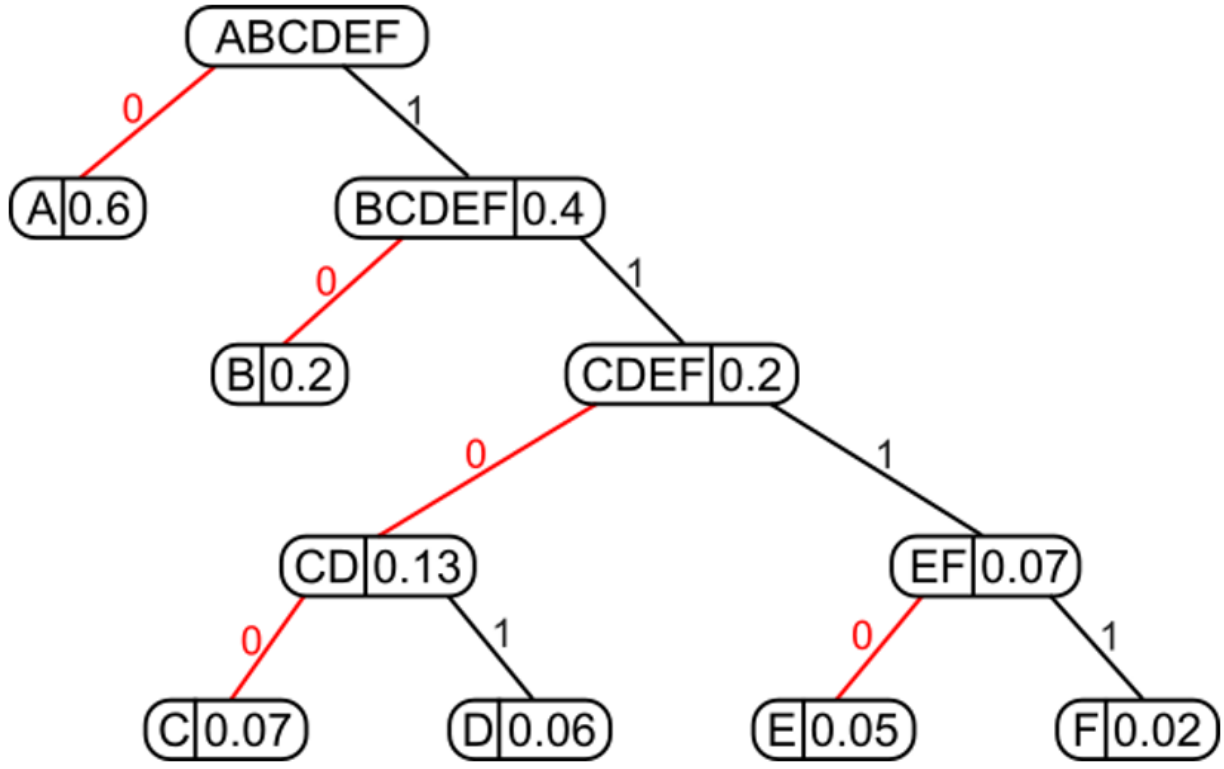


Figure 2.15 A Huffman coding tree built from this character frequency table:
A=0.6, B=0.2, C=0.07, D=0.06, E=0.05, F=0.02.